# Truth table to compound proposition

Given a truth table, how do we find an expression using the input variables and logical operators that has the output values specified in this table?

*Application*: design a circuit given a desired input-output relationship.

| Input | | Output | |
|---|---|---|---|
| $p$ | $q$ | $mystery_1$ | $mystery_2$ |
| $T$ | $T$ | $T$ | $F$ |
| $T$ | $F$ | $T$ | $F$ |
| $F$ | $T$ | $F$ | $F$ |
| $F$ | $F$ | $T$ | $T$ |

Expressions that have output $mystery_1$ are

Expressions that have output $mystery_2$ are

*Idea*: To develop an algorithm for translating truth tables to expressions, define a convenient **normal form** for expressions.

# Dnf cnf definition

**Definition** An expression built of variables and logical operators is in **disjunctive normal form** (DNF) means that it is an OR of ANDs of variables and their negations.

**Definition** An expression built of variables and logical operators is in **conjunctive normal form** (CNF) means that it is an AND of ORs of variables and their negations.

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